

Task 1 — Simulation and Sensitivity Analysis of Chaotic Systems

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1. Part A — Discrete Chaotic Systems

1.1 What is a discrete dynamical system?

A discrete system evolves in steps: the state at iteration $n+1$ is obtained by applying a fixed map to the state at iteration n , $x_{n+1} = f(x_n, \text{parameters})$. There is no time step, only an iteration count. The state variable is the quantity that describes the system at a given iteration (e.g. x for the logistic map, or the pair (x, y) for the Hénon map). The control parameter (e.g. r or a, b) is a fixed constant that is not part of the state but controls the qualitative behavior – fixed point, periodic, or chaotic – and can push the system through a cascade of bifurcations.

1.2 Logistic Map

The logistic map is a one-dimensional quadratic recurrence,

$$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$$

At $r = 3.9$ the map is in the chaotic regime: the trajectory never repeats and never settles onto a fixed point or a periodic orbit, and the return plot x_n vs x_{n+1} traces out the characteristic inverted parabola.

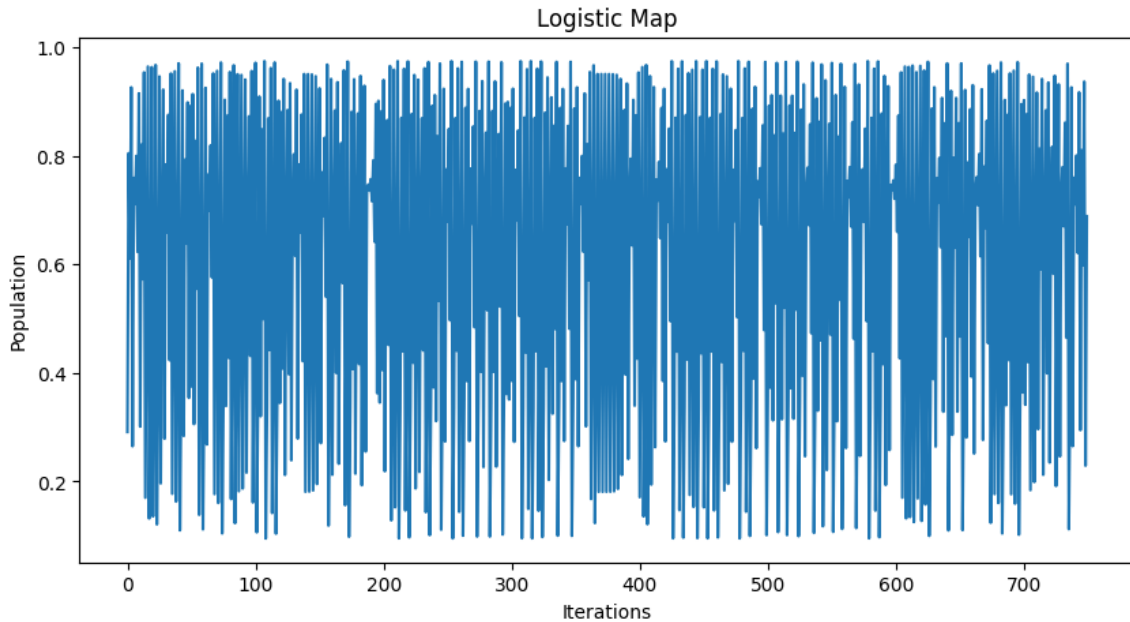


Figure A1.1 — Logistic map time series at $r = 3.9$ (after 250-iteration transient).

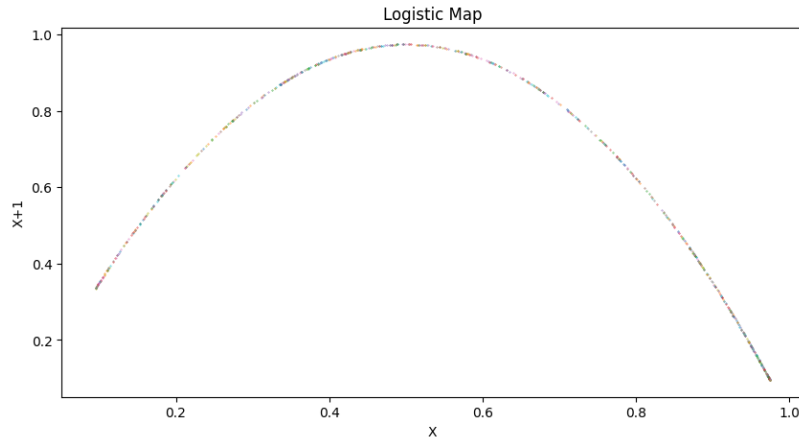


Figure A1.2 — Logistic return map: every iterate falls exactly on the parabola $r \cdot x \cdot (1 - x)$.

Settings:

Setting	Value
Equation	$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$
R	3.9
Initial condition x_0	0.5
Iterations	1000
Transient discarded	250

1.3 Hénon Map

The Hénon map is a two-dimensional generalisation of the logistic map:

$$x_{n+1} = 1 - a \cdot x_n^2 + y_n, \quad y_{n+1} = b \cdot x_n$$

At the standard values $a = 1.4$, $b = 0.3$ the iterates fill the well-known folded, banded attractor. The time series of x and y look erratic and non-periodic, and the (x, y) scatter reveals the fractal structure produced by the repeated stretch-and-fold action of the map.

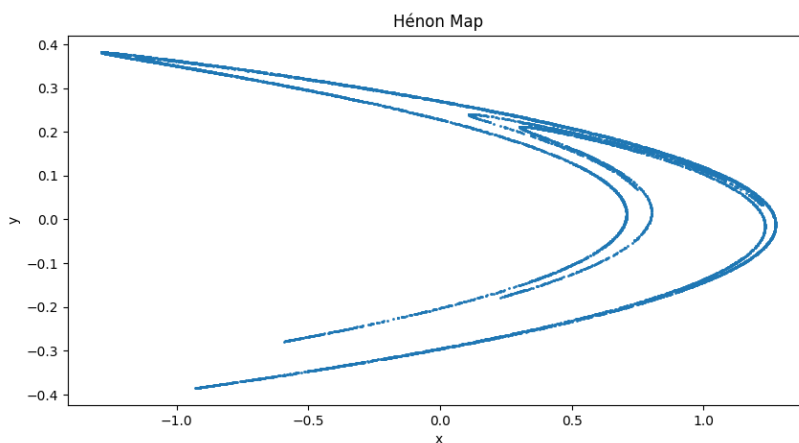


Figure A1.3 — Hénon attractor ($a = 1.4$, $b = 0.3$, 10 000 iterations after transient).

Settings:

Setting	Value
Equations	$x_{n+1} = 1 - a \cdot x_n^2 + y_n$; $y_{n+1} = b \cdot x_n$
a, b	1.4, 0.3
Initial condition (x_0, y_0)	(0, 0)
Iterations	10 000
Transient discarded	500

1.4 Bifurcation Diagrams

A bifurcation diagram shows how the long-term behaviour of a map changes as one control parameter is varied. For each parameter value the map is iterated many times, a transient is discarded, and the remaining iterates are plotted as points above that parameter value. The diagram exposes the transition from stable fixed points through period-doubling cascades to fully developed chaos.

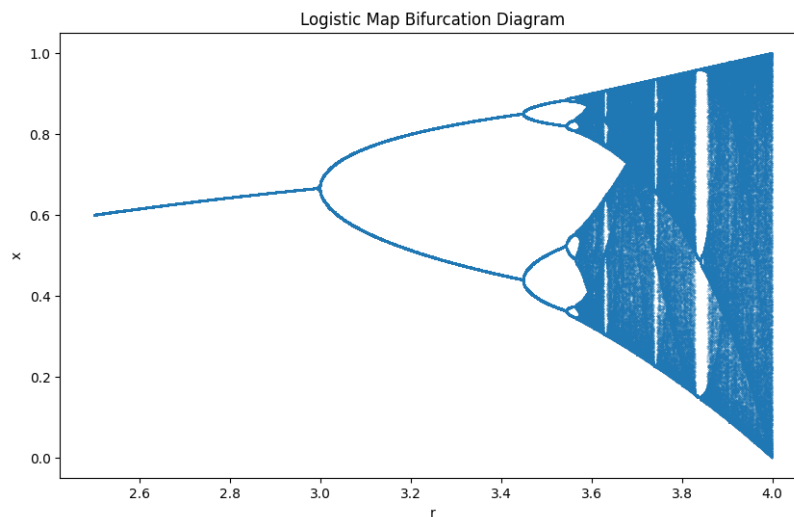


Figure A1.4 — Bifurcation diagram of the logistic map for $r \in [2.5, 4.0]$.

Observation. For $r < 3$ the map settles onto a single fixed point (one branch). At $r \approx 3$ a period-doubling cascade begins (period-2, period-4, period-8, ...), accumulating near $r \approx 3.5699$. Beyond this, the iterates fill dense chaotic bands, interrupted by narrow periodic windows — most visibly the period-3 window near $r \approx 3.83$.

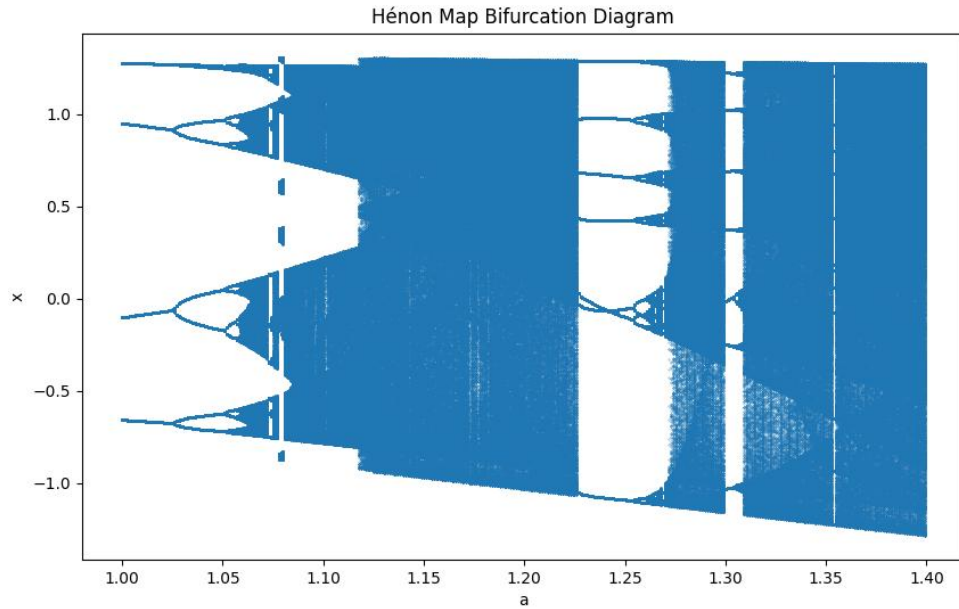


Figure A1.5 — Bifurcation diagram of the Hénon map ($b = 0.3$, a swept over $[1.0, 1.4]$).

Observation. The Hénon map shows the same qualitative route to chaos: a single-branch fixed point at small a splits via period-doubling into 2, 4, 8, ... branches, and then breaks up into wide chaotic bands. Narrow periodic windows are again visible inside the chaotic region.

Settings (both diagrams):

Setting	Value
Logistic sweep r	$[2.5, 4.0]$, step 0.001
Logistic iterations / transient	1000 / 500
Hénon sweep a ($b = 0.3$ fixed)	$[1.0, 1.4]$, step 0.001
Hénon iterations / transient	5000 / 500
Initial condition	$x_0 = 0.5$ (logistic); $(0, 0)$ (Hénon)

2. Part B — Continuous Chaotic Systems

2.2 Lorenz System

The Lorenz system, first proposed as a truncated model of atmospheric convection, is the prototypical continuous chaotic attractor:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = x \cdot y - \beta \cdot z$$

At the classical values $\sigma = 10$, $\rho = 28$, $\beta = 8/3$ the trajectory settles onto the famous butterfly-shaped attractor: the state variable spirals around one lobe, occasionally jumps unpredictably to the other, and never revisits the same point twice.

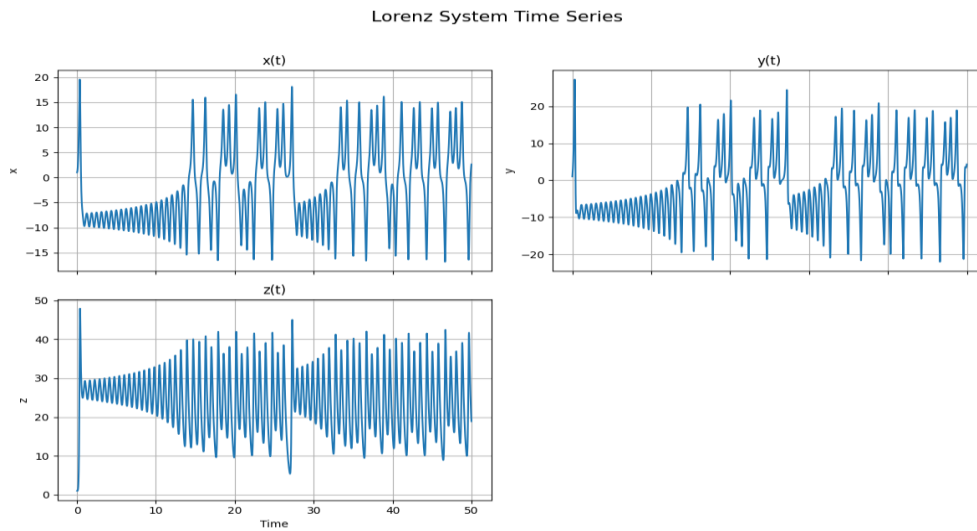


Figure B2.1 — Lorenz time series of x , y , z ($T = 50$).

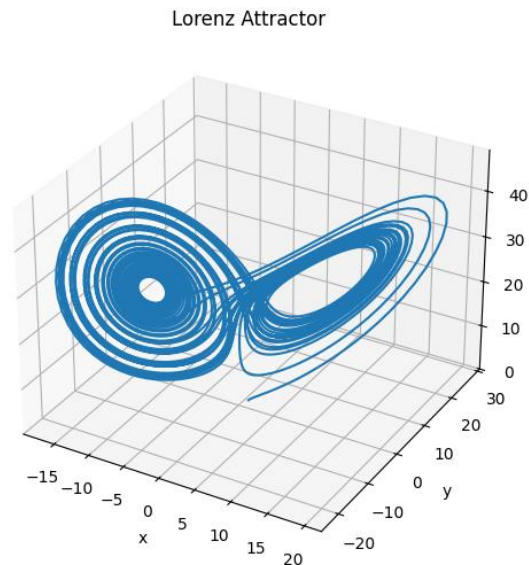


Figure B2.2 — Lorenz strange attractor in (x, y, z) phase space.

Settings:

Setting	Value
Parameters	$\sigma = 10, \rho = 28, \beta = 8/3$
Initial condition	(1, 1, 1)
Time span T	50
Output step dt	0.01
Solver	SciPy solve_ivp (RK45)

2.3 Rössler System

The Rössler system was designed as the simplest possible ODE system that exhibits chaos:

$$\dot{x} = -y - z, \quad \dot{y} = x + a \cdot y, \quad \dot{z} = b + z(x - c)$$

At $(a, b, c) = (0.2, 0.2, 5.7)$ the attractor is a nearly planar spiral with occasional rapid expansion in the z direction the trajectory winds outward, then folds back onto itself.

Rössler System

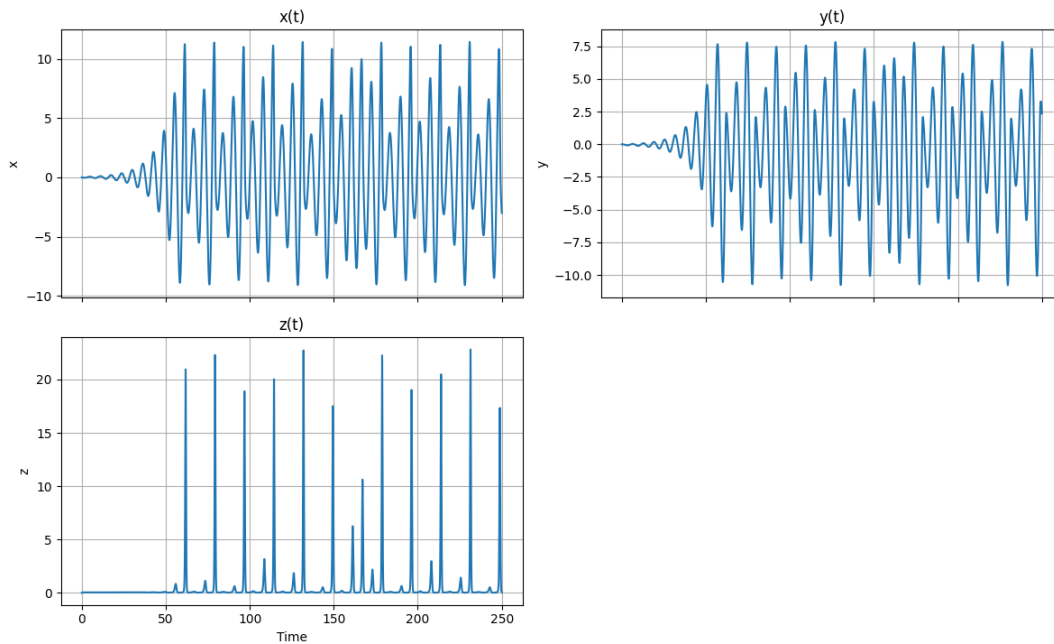


Figure B2.3 — Rössler time series ($T = 250$).

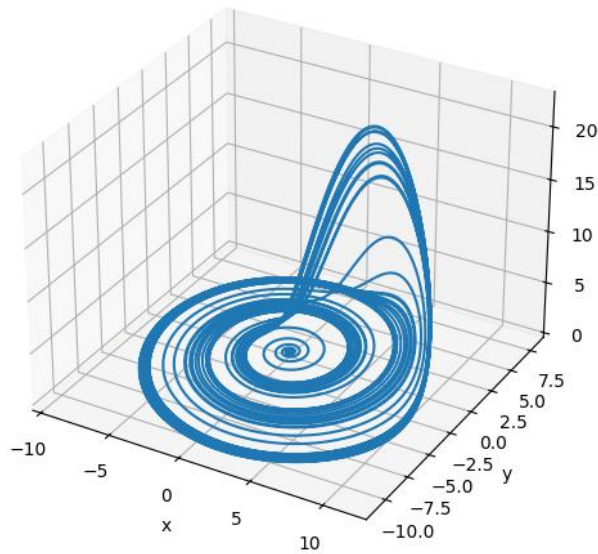


Figure B2.4 — Rössler strange attractor.

Settings:

Setting	Value
Parameters	$a = 0.2, b = 0.2, c = 5.7$
Initial condition	$(0, 0, 0)$
Time span T	250
Output step dt	0.01
Solver	SciPy solve_ivp (RK45)

2.4 Chen System

The Chen system, discovered by anti-controlling the Lorenz system, is topologically different from Lorenz but visually similar:

$$\dot{x} = a(y - x), \quad \dot{y} = (c - a)x - x \cdot z + c \cdot y, \quad \dot{z} = x \cdot y - b \cdot z$$

At $(a, b, c) = (35, 3, 28)$ the trajectory forms a two-lobed attractor like of Lorenz but with something like cone (lobe) in z-direction.

Chen System

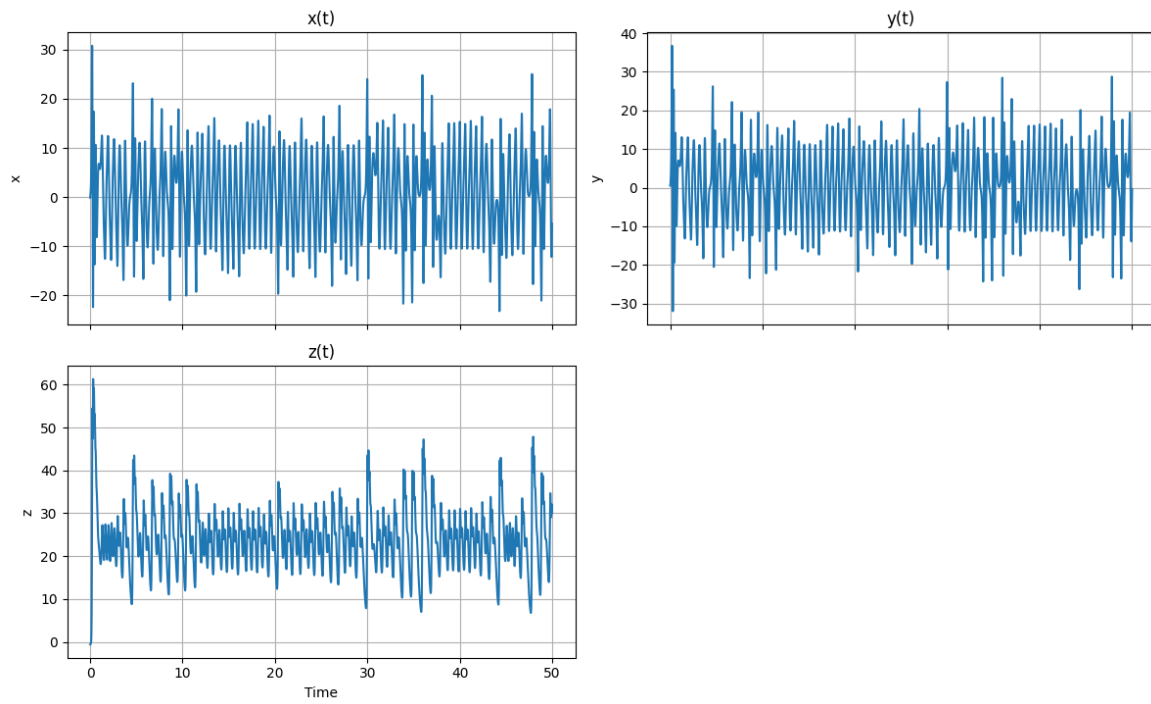


Figure B2.5 — Chen time series ($T = 50$).

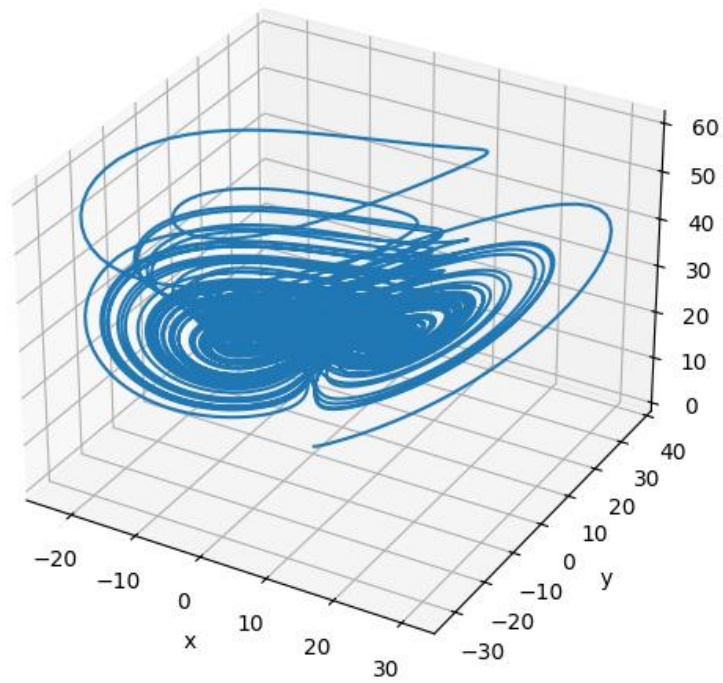


Figure B2.6 — Chen strange attractor.

Settings:

Setting	Value
Parameters	$a = 35, b = 3, c = 28$
Initial condition	$(-0.1, 0.5, -0.6)$
Time span T	50
Output step dt	0.003
Solver	SciPy solve_ivp (RK45)

2.5 Hyperchaotic Rössler System

Coupling a fourth state variable w into the Rössler system produces a 4-D flow with two positive Lyapunov exponents:

$$\dot{x} = -y - z, \quad \dot{y} = x + a \cdot y + w, \quad \dot{z} = b + x \cdot z, \quad \dot{w} = -c \cdot z + d \cdot w$$

Because the attractor lives in 4-D, we cannot draw it in a single 3-D plot; the six pairwise 2-D projections together characterise the geometry.

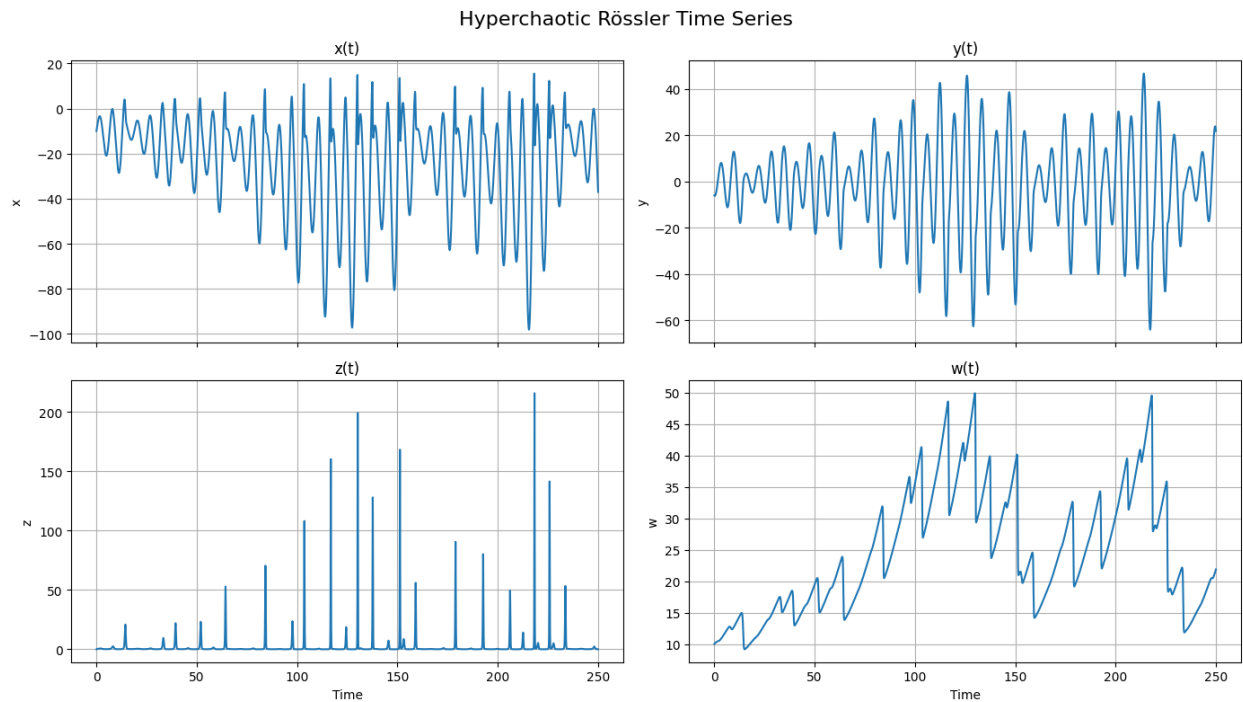


Figure B2.7 — Hyperchaotic Rössler time series of x, y, z, w ($T = 250$).

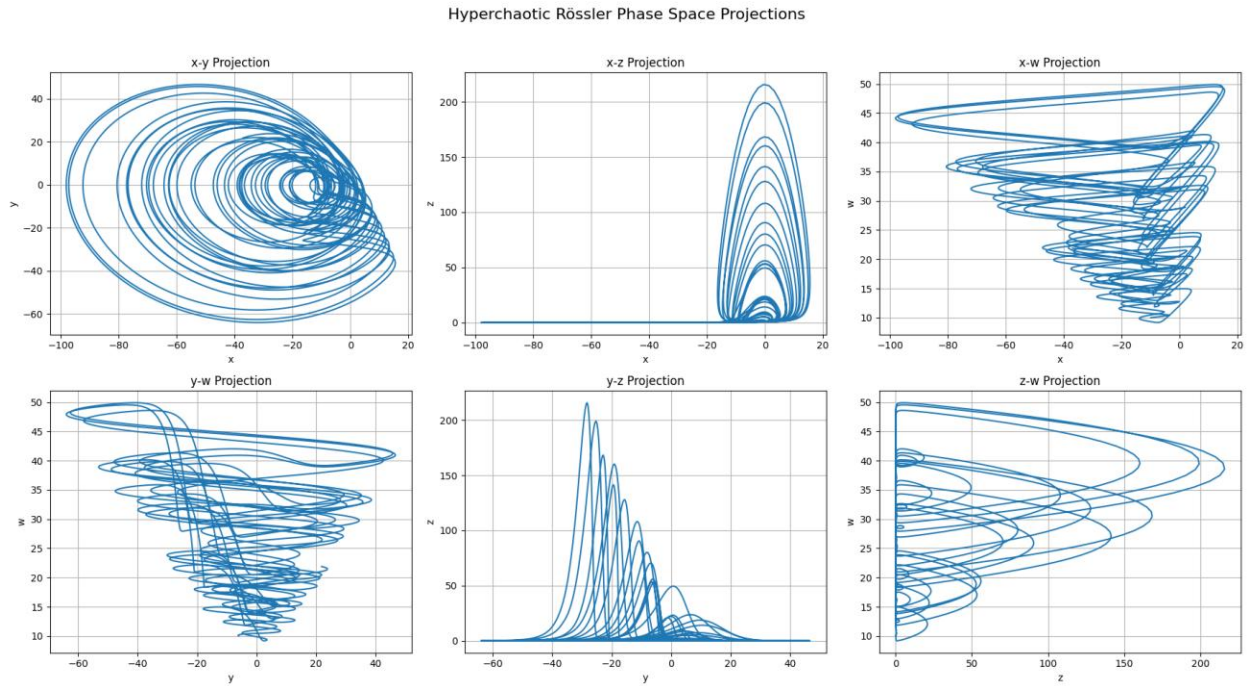


Figure B2.8 — Six 2-D projections of the 4-D hyperchaotic Rössler attractor.

Settings:

Setting	Value
Parameters	$a = 0.25, b = 3, c = 0.5, d = 0.05$
Initial condition	$(-10, -6, 0, 10)$
Time span T	250
Output step dt	0.01
Solver	SciPy solve_ivp (RK45)

3. Part C — Sensitivity Analysis

3.1 Parameter sweep of the Lorenz system (ρ)

Fixing $\sigma = 10$ and $\beta = 8/3$, ρ was swept over $[0, 30]$ in steps of 0.3. For each ρ the Lorenz system was integrated for $T = 50$, the first half of the trajectory was discarded as transient, and the local maxima of z were extracted and plotted. This is the continuous-system analogue of the discrete bifurcation diagrams in Part A.

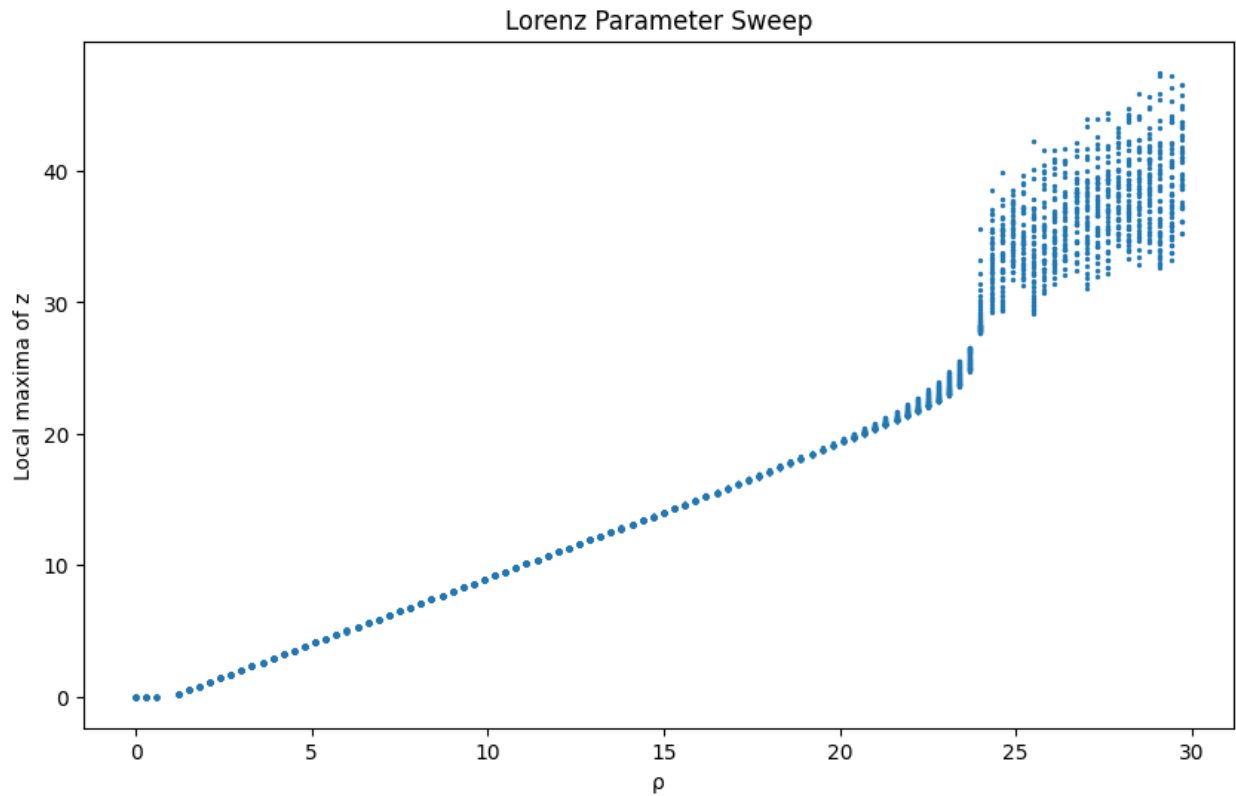


Figure C3.1 — Lorenz ρ -sweep: local maxima of z vs ρ ($\sigma = 10$, $\beta = 8/3$).

Observation. For $\rho \lesssim 1$ the origin is the only stable equilibrium — z relaxes to 0 and the plot shows a single point at zero. In the range roughly $1 \lesssim \rho < 25$ the system settles onto one of two stable fixed points, and the peak-of- z plot is a smooth single curve that grows with ρ . Just past $\rho \approx 25$ the fixed points lose stability and the trajectory jumps onto the chaotic butterfly attractor: the peak values suddenly spread over a wide band (roughly $z \in [30, 45]$), signalling the onset of chaos.

3.2 Step-size sensitivity

The Lorenz system was re-simulated at the standard chaotic parameters ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) with the same solver, initial condition and total simulation length ($T = 50$), using three different step sizes $dt \in \{0.001, 0.01, 0.05\}$. Note that dt here is the `max_step` passed to `solve_ivp`, so a large dt does force the adaptive solver into cruder steps because `solve_ivp` uses adaptive learning rate and we can't change step directly

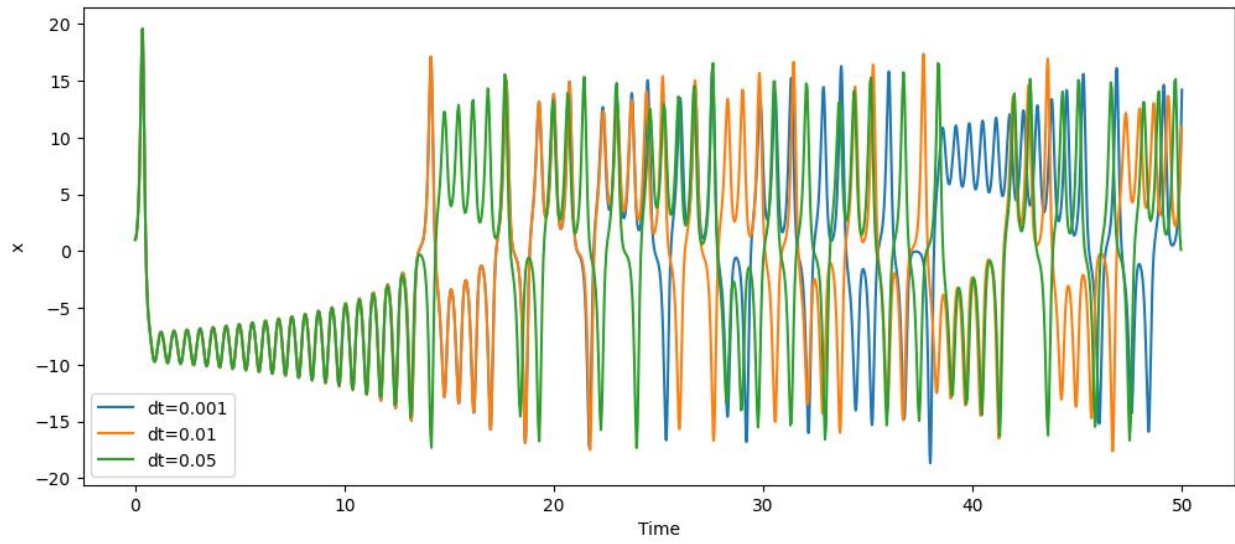


Figure C3.2 — Lorenz $x(t)$ at three step sizes.

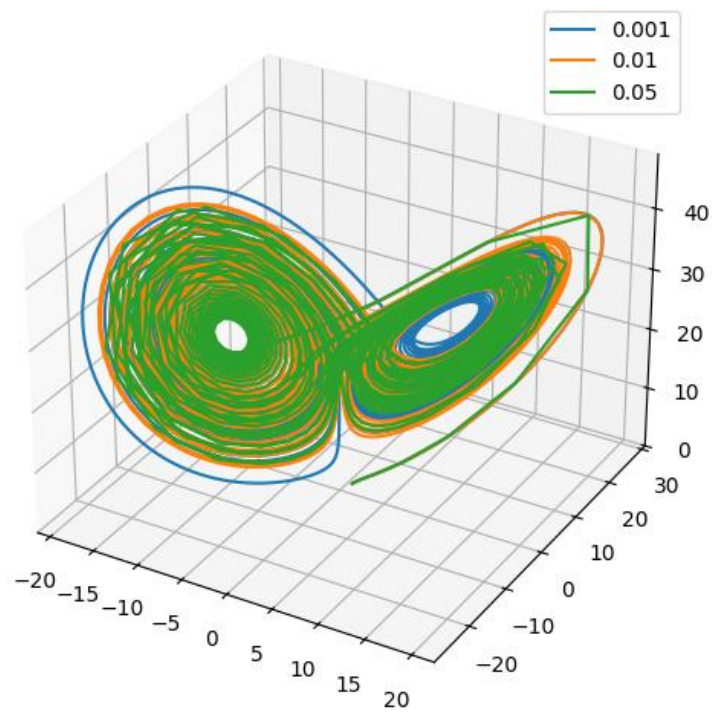


Figure C3.3 — 3-D attractor overlay for the three step sizes.

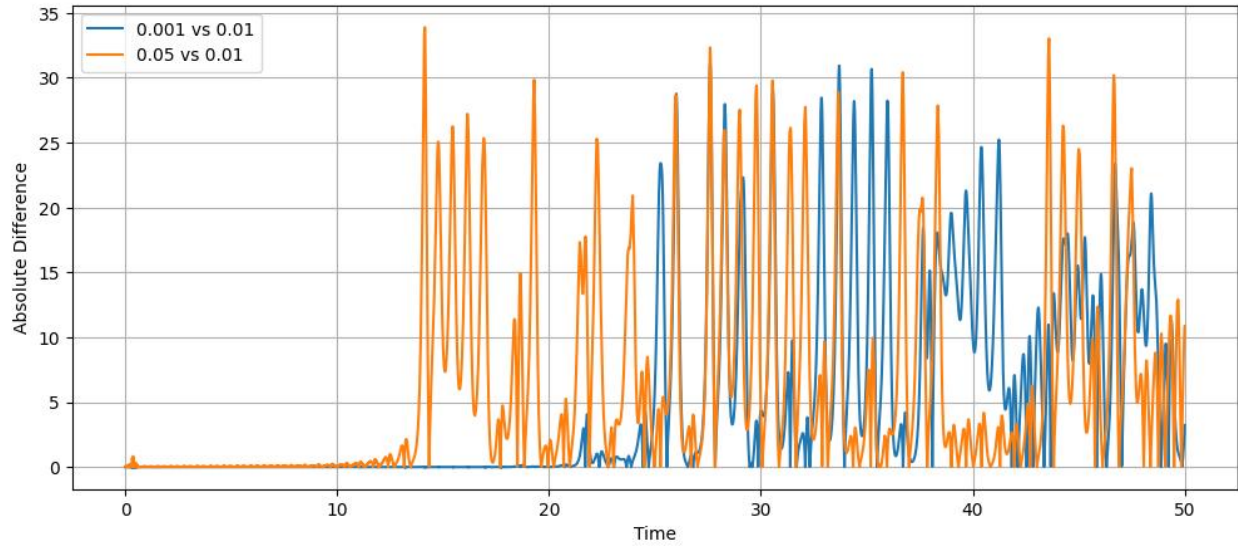


Figure C3.4 — Absolute difference $|\Delta x|$ between each simulation and the $dt = 0.01$ baseline.

Observation. For roughly the first 10 times units all three trajectories are visually indistinguishable. Beyond that, the finest solution ($dt = 0.001$) begins to drift away from the $dt = 0.01$ baseline, and by $t \approx 25$ – 30 the two solutions lie on different lobes of the attractor. The coarse solution ($dt = 0.05$) diverges from the baseline much sooner (around $t \approx 10$). This is sensitive to initial conditions: the tiny numerical differences introduced by different step sizes are amplified exponentially by the positive Lyapunov exponent, so individual trajectories diverge even though all three trajectories remain on the same attractor (as the 3-D overlay confirms).